

The frustum of a cone

The bucket, the lampshade and the flowerpot — and the two formulas that measure them.

LESSON 39

1 × 40 MIN

GEOMETRY 11, §18

EXTENSION

LESSON CLOCK

8'

14'

14'

6'

3'

The solid	8 min
Surface area	14 min
Volume	14 min
Practical problems	6 min
Homework	3 min

LEARNING OBJECTIVES

- Describe the elements of a frustum of a cone.
- Compute its curved and total surface area.
- Compute its volume.
- Solve a practical problem about a bucket or a lampshade.

TEXTBOOK

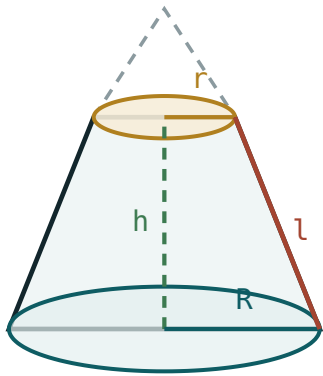
National	pp. 195–200
Cambridge	Core & Extended, pp. 343–346
Workbook	Exercise 18.1

01

Explanation

The solid

Cut a cone with a plane parallel to the base and remove the small cone. What remains is a **frustum of a cone** — the shape of every bucket, lampshade and flowerpot.



$$V = \frac{1}{3}\pi h(R^2 + Rr + r^2) \cdot S_{\text{lat}} = \pi l(R + r)$$

The removed cone shown dashed. R and r are the two radii; l is the slant height of the frustum.

$$l^2 = h^2 + (R - r)^2$$

The axial section is a trapezium with parallel sides $2R$ and $2r$, height h and slanting sides l . The $R - r$ in the formula is the horizontal step from one radius to the other.

Surface area

$$S_{\text{curved}} = \pi l(R + r) \quad S_{\text{total}} = \pi l(R + r) + \pi R^2 + \pi r^2$$

It is the big cone's curved area minus the small one's, simplified — the same construction as the pyramid frustum, with $\frac{1}{2}(P_1 + P_2)l$ becoming $\pi(R + r)l$.

A BUCKET HAS ONE END, NOT TWO

A bucket needs $\pi l(R + r) + \pi r^2$ — the curved surface and the **base** only. Read the object before choosing which circles to include.

Volume

$$V = \frac{\pi h}{3}(R^2 + Rr + r^2)$$

The same shape of formula as the pyramid frustum, with $\sqrt{B_1 B_2}$ becoming πRr . With $r = R$ it gives the cylinder $\pi R^2 h$; with $r = 0$ it gives the cone $\frac{1}{3}\pi R^2 h$. Both checks are worth doing once.

Example. A bucket of radii 12 and 9 cm and depth 20 cm:

$$V = \frac{20\pi}{3}(144 + 108 + 81) = \frac{20\pi}{3}(333) = 2220\pi \approx 6975 \text{ cm}^3$$

That is about 7 litres — a realistic bucket.

Practical problems

OBJECT	WHAT IS WANTED	WHICH FORMULA
a bucket	capacity	$\frac{\pi h}{3}(R^2 + Rr + r^2)$
a lampshade	material	$\pi l(R + r)$
a flowerpot	capacity and the material for the sides and base	both
a paper cup	the flat sector-ring to cut	the net

THE CHECK THAT CATCHES ERRORS

Put $r = R$ into any frustum formula: it must give the cylinder. Put $r = 0$: it must give the cone. A formula that fails either check has been mis-copied.

02 Worked examples

EXAMPLE 1 A bucket has radii 12 and 9 cm and depth 20 cm. Find its capacity in litres.

1 $V = \frac{20\pi}{3}(144 + 108 + 81)$

2 $= \frac{20\pi}{3} \times 333 = 2220\pi$

3 $\approx 6975 \text{ cm}^3$

4 $\approx 7.0 \text{ litres.}$

ANSWER

$\approx 7.0 \text{ litres}$

EXAMPLE 2 The same bucket: find the area of metal needed (curved surface and base).

$$① \quad l = \sqrt{400 + 9} = \sqrt{409} \approx 20.2$$

$$② \quad S_{\text{curved}} = \pi(20.2)(21) \approx 1333$$

$$③ \quad \text{base} = \pi(81) \approx 254$$

$$④ \quad \approx 1587 \text{ cm}^2.$$

ANSWER

$$\approx 1590 \text{ cm}^2$$

EXAMPLE 3 A frustum has radii 6 and 3 and slant height 5. Find its height and volume.

$$① \quad h = \sqrt{25 - 9} = 4$$

$$② \quad V = \frac{4\pi}{3}(36 + 18 + 9)$$

$$③ \quad = \frac{4\pi}{3} \times 63 = 84\pi$$

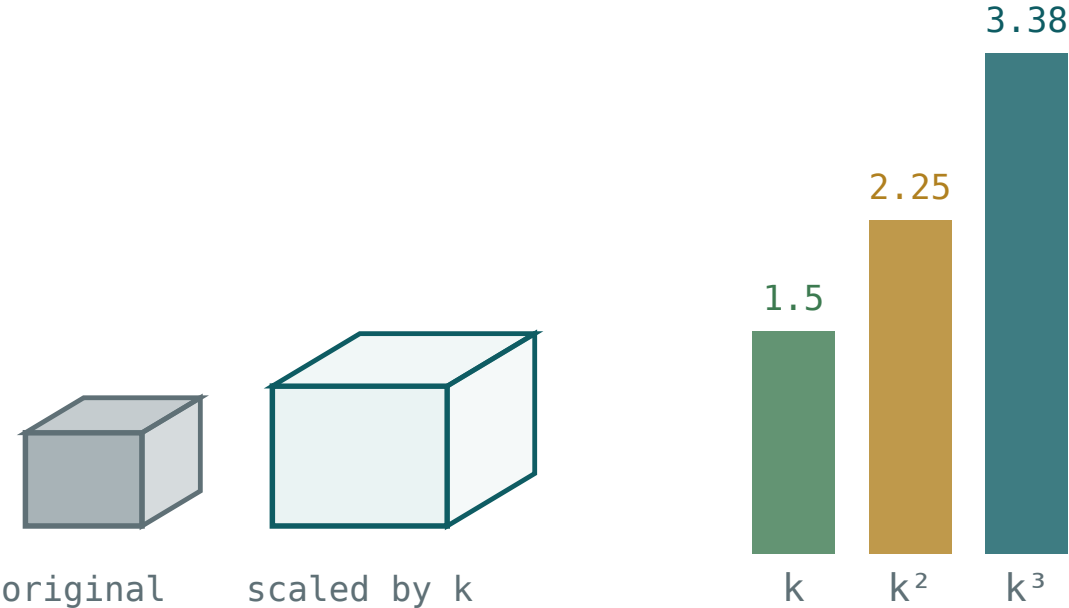
ANSWER

$$h = 4, V = 84\pi \approx 264$$

03 Interactive model

Bring a real bucket and measure its two radii and depth.

Cutting a cone



k 1.5	length × 1.5	area ×k² 2.25	volume ×k³ 3.38
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04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Frustum of a cone	Kesik konus	Усечённый конус
Upper radius	Yuqori radius	Верхний радиус
Lower radius	Quyi radius	Нижний радиус
Slant height	Yasovchi	Образующая
Curved surface	Yon sirt	Боковая поверхность
Capacity	Sig'im	Вместимость
Complementary cone	To'ldiruvchi konus	Дополняющий конус
Axial section	O'qli kesim	Осевое сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself l^2 for a frustum equals:

$$h^2 + R^2$$

$$h^2 + (R - r)^2$$

$$h^2 + (R + r)^2$$

$$R^2 - r^2$$

 S_{curved} of a frustum:

$$\pi r l$$

$$\pi l(R + r)$$

$$\pi l(R - r)$$

$$2\pi r l$$

 V of a frustum:

$$\frac{\pi h}{2}(R^2 + r^2)$$

$$\frac{\pi h}{3}(R^2 + Rr + r^2)$$

$$\frac{1}{3}\pi R^2 h$$

$$\pi R^2 h$$

Setting $r = R$ gives:

a cone

a cylinder

a sphere

nothing

Setting $r = 0$ gives:

a cone

a cylinder

a sphere

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. l with $h = 4$, $R = 6$, $r = 3$

ANSWER 5

2. l with $h = 12$, $R = 10$, $r = 5$

ANSWER 13

3. S_{curved} with $l = 5$, $R = 6$, $r = 3$

ANSWER 45π

4. V with $h = 4$, $R = 6$, $r = 3$

ANSWER 84π

5. Setting $r = R$ gives

ANSWER a cylinder

6. Setting $r = 0$ gives

ANSWER a cone

7. Axial section of a frustum

ANSWER a trapezium

Medium · the standard question

7 PROBLEMS

1. Bucket $R = 12, r = 9, h = 20$: capacity in litres

ANSWER ≈ 7.0

2. Same: metal for the sides and base

ANSWER $\approx 1590 \text{ cm}^2$

3. $R = 10, r = 4, h = 8$: the volume

ANSWER $\frac{8\pi}{3} \times 156 = 416\pi$

4. Same: the slant height

ANSWER 10

5. Same: the curved area

ANSWER 140π

6. A lampshade $R = 20, r = 12, h = 15$: material

ANSWER $l = 17$, curved area 544π

7. A flowerpot $R = 15, r = 10, h = 18$: capacity in litres

ANSWER ≈ 8.9

Hard · stretch and reason

7 PROBLEMS

1. A frustum of volume 84π with $R = 6$, $r = 3$: its height

ANSWER 4

2. A bucket holds 10 litres with $R = 14$, $r = 11$: its depth

ANSWER ≈ 20.6 cm

3. Show V reduces to the cone when $r = 0$

ANSWER The other two terms vanish

4. A frustum cut halfway up a cone: the volume ratio frustum : whole

ANSWER 7 : 8

5. A frustum with $R = 2r$ and height h : its volume in terms of r , h

ANSWER $\frac{7\pi r^2 h}{3}$

6. A paper cup $R = 4$, $r = 2.5$, $h = 9$: capacity in ml

ANSWER ≈ 313

7. Two similar frustums with heights 3 and 5: the volume ratio

ANSWER 27 : 125

07 Homework

Homework — 4 tasks

Check every formula by setting $r = R$ and $r = 0$.

1. A bucket has radii 14 and 10 cm and depth 24 cm. Find its capacity in litres.
2. The same bucket: find the area of metal for the curved surface and the base.
3. A frustum has radii 8 and 5 and slant height 5. Find its height, curved area and volume.
4. Show that the frustum volume formula gives the cylinder when $r = R$ and the cone when $r = 0$.